

TACHS Exam Simulator

Full-Length Practice Exam 02

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: The Bell in the Box

Maya had promised to help her grandfather sort the contents of his workshop before the new shelves arrived. Most objects were easy to classify: screws went into jars, sandpaper into a drawer, and rulers into a wooden bin. Then she found a small brass bell wrapped in newspaper.

"That belonged above the door of my first bicycle shop," Grandfather said. He tapped it, but the bell produced only a dull click.

Maya examined the clapper and discovered that a thin wire had bent against the side. She reached for pliers, then stopped. "If I straighten it too quickly, it might snap."

Her grandfather nodded and returned to labeling boxes. Maya loosened the wire a little at a time, testing the bell after each adjustment. At last, a clear note floated through the dusty room. Grandfather looked up, smiling.

They hung the bell beside the workshop door. During the following week, Maya noticed that her grandfather rang it whenever he entered, even though it no longer announced a customer. On the day the shelves arrived, he handed her the shop's faded wooden sign. "You understood that some things need patience, not replacement," he said. "You should decide where this goes."

1. What caused the bell to make only a dull click?

- A) Dust had filled the bell.
- B) A bent wire pressed against its side.
- C) The clapper was missing.
- D) The brass had cracked.

2. Why does Grandfather probably ring the bell when entering the workshop?

- A) He wants Maya to know that he needs help.
- B) He is checking whether the repair will last.
- C) The sound connects him with memories of his first shop.
- D) The shelves make it difficult for him to see the door.

3. Which statement best expresses a central idea of the passage?

- A) Careful attention can preserve objects and the memories they carry.
- B) Organizing a workshop requires more time than expected.
- C) Old business signs are more valuable than new furniture.
- D) Repair work should always be done with specialized tools.

4. As used in the passage, "floated" most nearly means

- A) rested on a liquid.
- B) moved without direction.
- C) remained suspended.
- D) traveled gently.

Informational Passage: Why Some Bridges Hum

On windy days, a bridge may produce a low hum, a whistle, or even a musical tone. These sounds do not necessarily indicate danger. They often begin when moving air interacts with railings, cables, or narrow gaps in the bridge's structure.

As wind passes an edge, it can break into repeating swirls. Each swirl briefly changes the air pressure nearby. If those pressure changes occur at a steady rate, they create a vibration that people hear as a tone. The effect resembles the sound made by blowing across the opening of a bottle, although a bridge offers many more surfaces for the air to meet.

Engineers study these sounds because vibration matters even when a structure is safe. A persistent tone may disturb nearby residents, and repeated vibration can loosen small parts over time. Engineers may add thin barriers, alter the shape of a railing, or cover a gap so that air no longer forms regular swirls.

The goal is not always to silence the bridge completely. Wind changes direction and speed, so no single modification prevents every sound. Instead, engineers identify the surfaces producing the strongest vibrations and reduce those effects. Listening, therefore, becomes one of several tools for understanding how a bridge responds to its environment.

5. According to the passage, what directly creates a tone near a bridge edge?

- A) Steady changes in air pressure caused by repeating swirls
- B) Vehicles crossing narrow gaps at regular intervals
- C) Rainwater collecting inside hollow railings
- D) Cables striking one another in strong wind

6. Why does the author compare a humming bridge to blowing across a bottle?

- A) To argue that bridges should be designed like instruments
- B) To show that bottles create louder sounds than bridges
- C) To warn that trapped air can break bridge surfaces
- D) To explain an unfamiliar process through a familiar example

7. How is the passage mainly organized?

- A) It lists bridge designs from oldest to newest.
- B) It explains a cause, describes its effects, and presents responses.
- C) It compares two competing theories about bridge safety.
- D) It gives directions for building a model bridge.

8. As used in the third paragraph, "persistent" most nearly means

- A) dangerous.
- B) unexpected.
- C) continuing.
- D) artificial.

Humanities Passage: Reading the Empty Space

When people first study a drawing, they often concentrate on the objects: a vase, a face, or a row of buildings. Artists, however, also pay close attention to the unoccupied areas around and between those objects. This “negative space” is not simply a blank background. Its shape helps determine how the entire composition feels. Consider a poster crowded with words and pictures. Even if every element is interesting, the viewer may not know where to look first. Empty areas can separate groups of information and guide the eye from a heading to an image and then to a smaller line of text. In a portrait, the open area beside a person’s face may create a sense of quiet, while a figure pressed close to every edge can seem confined.

Negative space also assists artists in observing accurately. Instead of trying to draw the complicated outline of a chair, a student might draw the triangular gaps between its legs. When those gaps have the correct proportions, the chair itself often becomes more accurate.

Paying attention to what is absent may feel unnatural at first. Yet this shift does not make the subject less important. It reveals that objects and their surroundings work together. A successful composition depends not only on what an artist includes, but also on the room each element is allowed to occupy.

9. What is the passage mainly about?

- A) Why beginning artists should avoid drawing crowded scenes
- B) How poster designers select words and pictures
- C) Why accurately drawing furniture requires geometry
- D) How empty areas contribute to observation and composition

10. According to the passage, how can negative space improve a poster?

- A) It can organize information and guide the viewer’s eye.
- B) It can make every picture appear the same size.
- C) It can remove the need for a clear heading.
- D) It can allow more words to fit on the page.

11. What can reasonably be inferred about drawing the gaps between chair legs?

- A) It is useful only when the chair has a simple design.
- B) It prevents students from seeing the chair as a whole.
- C) It provides an indirect way to judge the chair’s proportions.
- D) It allows students to complete a drawing without observing carefully.

12. The author’s tone toward the study of negative space is best described as

- A) doubtful and guarded.
- B) explanatory and encouraging.
- C) humorous and playful.
- D) critical and impatient.

13. Which principle would the author most likely support?

- A) The relationship between elements can matter as much as the elements themselves.
- B) A realistic subject always produces the strongest composition.
- C) Empty areas should be placed only along the edges of a work.
- D) Artists should plan backgrounds before choosing a subject.

Literary or Narrative Passage: The Borrowed Route

Nia had planned every turn of the neighborhood bicycle tour, but ten minutes before the riders arrived, orange barriers blocked Willow Street. A water crew had opened a trench across the road. Nia unfolded her map on a bench and stared at the neat blue line she had drawn the night before.

"We could cancel," said Mr. Lee, the community-center director.

Nia shook her head. Several younger riders had practiced for weeks. She walked to the corner and studied the side streets. Juniper Lane curved toward the park, but she had never ridden its entire length. Just then, Mrs. Ortiz rolled up with a basket of library books on her bicycle. She explained that Juniper ended at a short gravel path before rejoining the planned route.

Nia tested the path herself. It was firm, wide, and free of traffic, though riders would need to walk their bicycles for fifty yards. She returned, marked the change, and explained it clearly at the safety meeting. During the tour, the group moved slowly over the gravel. The delay allowed them to notice a community garden hidden behind tall hedges. On the ride back, a child asked whether next month's route could include the garden on purpose.

Nia smiled as she redrew the line on her map. The detour no longer looked like a mistake.

14. What did Nia do before choosing Juniper Lane as the detour?

- A) She tested the gravel path herself.
- B) She asked the riders to vote.
- C) She moved the safety meeting.
- D) She called the water crew.

15. Why does Nia shake her head when Mr. Lee suggests canceling?

- A) She believes the barriers will soon be removed.
- B) She feels responsible to the riders who prepared.
- C) She wants Mrs. Ortiz to lead the tour.
- D) She is unwilling to change her map.

16. Which statement best expresses the central idea of the passage?

- A) Familiar routes are always safer than new ones.
- B) Community events require perfect weather and planning.
- C) An unexpected obstacle can lead to a worthwhile discovery.
- D) Leaders should avoid making decisions without a vote.

17. As used in the passage, "firm" most nearly means

- A) strict.
- B) determined.
- C) tightly held.
- D) solid enough to travel over.

Scientific or Technical Passage: A Seed's Spiral Descent

Some maple seeds travel through the air like tiny helicopters. Each seed is attached to a thin wing. When the seed falls, air pushes unevenly against the wing and begins turning it. This rotation, called autorotation, slows the seed's downward motion.

The spinning wing creates differences in air pressure. Air moves faster over one part of the wing than another, producing an upward force. That force does not lift the seed back into the tree, but it reduces the speed of its fall. A slower descent gives wind more time to carry the seed away from the parent tree.

Researchers study this motion by dropping seed models in still air and recording them with high-speed cameras. They may change one feature at a time, such as wing length, seed mass, or the angle at which the wing bends. Comparing the recordings helps them determine how each feature affects rotation and falling speed.

This research has uses beyond explaining how trees spread. Engineers have tested small spinning devices that descend without motors or parachutes. Such devices might carry lightweight sensors through the air to measure temperature or air quality over a broad area. The maple seed offers a model: a simple shape can control motion by working with the surrounding air.

18. According to the passage, what is one effect of a maple seed's rotation?

- A) It increases the seed's mass.
- B) It causes the wing to grow longer.
- C) It returns the seed to the tree.
- D) It slows the seed's downward motion.

19. Why does the author mention high-speed cameras?

- A) To show that seed models are difficult to construct
- B) To explain how researchers observe and compare rapid motion
- C) To argue that natural seeds are unsuitable for experiments
- D) To describe how sensors record air quality

20. How is the passage mainly organized?

- A) It explains a natural process, describes how it is studied, and identifies a possible application.
- B) It compares two tree species and ranks their methods of spreading seeds.
- C) It presents a problem, lists failed solutions, and recommends one experiment.
- D) It tells the history of a discovery in chronological order.

21. As used in the final paragraph, "model" most nearly means

- A) a small copy made only for display.
- B) a person who demonstrates a product.
- C) an example that can guide a design.
- D) a calculation that predicts exact weather.

Informational or Historical-Style Passage: When Towns Numbered Their Doors

In many growing towns, street addresses were not always part of daily life. Residents might direct a visitor to “the blue house beyond the mill” or identify a shop by its owner’s name. Such descriptions worked when communities were small and landmarks were widely known.

As towns expanded, informal directions became less reliable. New residents did not recognize every family name, and several streets might contain similar buildings. Postal workers, firefighters, and delivery drivers needed a system that allowed them to locate a property without knowing its history.

Town officials began assigning names to streets and numbers to buildings. A useful numbering plan followed a predictable order. Often, odd numbers appeared on one side of a street and even numbers on the other. In some places, numbers increased with distance from a central road or town square. Signs at intersections made the system visible to people passing through.

Adopting addresses required more than attaching plaques to doors. Officials had to create accurate records, correct duplicate street names, and inform residents of changes. Despite the work involved, standardized addresses gradually became an important layer of public information. They helped strangers navigate a place by following shared rules rather than depending on local memory.

22. Which groups does the passage specifically identify as needing a dependable location system?

- A) Builders, teachers, and shop owners
- B) Visitors, mill workers, and officials
- C) Postal workers, firefighters, and delivery drivers
- D) Farmers, surveyors, and police officers

23. What is the author’s main purpose in the passage?

- A) To explain why standardized addresses became necessary and how they improved navigation
- B) To persuade towns to replace numbers with landmark descriptions
- C) To criticize residents who resisted changing street names
- D) To compare the architecture of small and large towns

24. Which statement best summarizes the passage?

- A) Street signs were mainly created to decorate intersections in growing towns.
- B) Address systems replaced increasingly unreliable local directions with predictable, shared location information.
- C) Postal workers designed identical address systems for every town.
- D) Duplicate street names disappeared as soon as buildings received plaques.

25. Which inference is best supported by the passage?

- A) Every resident welcomed standardized addresses immediately.
- B) Building numbers were assigned before towns began to grow.
- C) Landmark directions remained equally useful to newcomers.
- D) A successful address system depends on consistent records as well as visible numbers.

Literary Passage: The Last Lantern

Each autumn, Lena's apartment building held a lantern walk in the courtyard. Families carried glowing paper lanterns while Mr. Alvarez played old songs on his accordion. This year, however, the event nearly disappeared. The courtyard was being repaired, and the adults decided there was nowhere suitable to gather. Lena disliked accepting that answer. In the basement, she noticed a long hallway leading past the storage rooms. It was plain and narrow, but it was also dry and free of construction equipment. She measured the hallway, counted the working outlets, and asked the superintendent whether the space could be used safely. With permission, Lena invited neighbors to hang lanterns from removable hooks. Children painted paper panels with scenes from every floor of the building: Mrs. Chen's window garden, the twins' orange cat, and the rooftop water tank. Because an accordion would echo too loudly underground, Mr. Alvarez played softly near the entrance. The walk was shorter than usual, yet people moved slowly, recognizing familiar details in the pictures. At the end of the hallway, Lena found one unpainted lantern. She added a picture of the basement door standing open. It was not the courtyard they remembered, but the narrow passage had become a map of the community above it.

26. What did Lena do before asking to use the basement hallway?

- A) She measured the space and checked the outlets.
- B) She painted a picture of the basement door.
- C) She moved the construction equipment.
- D) She invited Mr. Alvarez to perform.

27. Why do people move slowly through the hallway?

- A) They are concerned that the lanterns may fall.
- B) They enjoy identifying details from their shared surroundings.
- C) They are waiting for the accordion music to begin.
- D) They have difficulty seeing the basement door.

28. Which statement best expresses a central idea of the passage?

- A) Annual traditions should always remain unchanged.
- B) Apartment buildings need large spaces for community events.
- C) Creativity can preserve a tradition by giving it a new form.
- D) Children notice more neighborhood details than adults do.

29. As used in the passage, "plain" most nearly means

- A) obvious.
- B) truthful.
- C) level.
- D) undecorated.

Literary Passage: A Measure of Rain

Owen had spent three weeks building a wooden rain gauge for the school garden. He sanded the post, marked a clear container in centimeters, and sealed every joint. On the morning of the science fair, he carried it outside and discovered that the container was empty, although rain had fallen all night.

His first thought was to hide the gauge behind the tomato plants. Then he noticed a thin trail of water running down the post. The seal around one screw had failed. Owen's display board promised "accurate daily rainfall," and visitors would arrive within an hour.

He found his teacher, Ms. Patel, and explained the leak. Together they placed a new container beside the faulty one. Owen added a card describing what had happened and invited visitors to predict whether the repair would work. Throughout the fair, students inspected the stain on the post and suggested ways to improve the seal.

Near closing time, a brief shower crossed the garden. The new container collected four millimeters of water. Owen was relieved, but he left the leaking gauge in place. His project had not produced the week of measurements he expected. Instead, it showed why scientists record problems rather than erase them. The empty container had become evidence too.

30. What caused the original rain gauge to lose water?

- A) The seal around a screw failed.
- B) The measurement marks washed away.
- C) The container tipped during the night.
- D) The tomato plants blocked the rain.

31. Why does Owen leave the leaking gauge on display?

- A) He plans to collect water from both containers.
- B) It helps demonstrate the problem and the investigation that followed.
- C) Ms. Patel requires every original model to remain outdoors.
- D) He believes visitors will not notice the leak.

32. The author's attitude toward Owen's response to failure is best described as

- A) amused by his unnecessary concern.
- B) doubtful that his repair will last.
- C) approving of his honesty and willingness to learn.
- D) disappointed by his lack of rainfall records.

33. As used in the final sentence, "evidence" most nearly means

- A) a prize for successful work.
- B) an opinion about an event.
- C) a replacement for measurement.
- D) information that helps explain what occurred.

Informational Passage: How Libraries Protect Quiet

Quiet rooms do not depend only on people whispering. Their design can also limit how sound travels. In a library, sound may bounce from hard floors, glass walls, and bare ceilings. These reflections allow a chair scraping in one corner to be heard far away.

Designers reduce reflected sound by including materials that absorb some of its energy. Carpet, fabric-covered panels, and certain ceiling tiles contain tiny spaces where vibrating air loses energy. Soft materials are not placed everywhere, however. Bookshelves already interrupt sound paths, and durable flooring may be necessary near entrances where rain and dirt are common.

The arrangement of rooms matters as well. Group-study areas can be separated from silent reading spaces by offices, stairways, or rows of shelving. Doors with tight seals prevent sound from escaping through gaps. Even the location of a return desk can make a difference because dropped books create brief, sharp noises.

No design can remove every sound, nor should it. People must communicate, alarms must be heard, and public spaces need to feel welcoming. Effective acoustic design therefore aims for balance. By controlling reflections and separating activities, a library can provide both lively places for cooperation and calmer places for concentration.

34. According to the passage, why might durable flooring be used near library entrances?

- A) It can withstand the rain and dirt brought inside.
- B) It absorbs more sound than ceiling tiles.
- C) It makes dropped books easier to hear.
- D) It directs visitors toward study areas.

35. What is the passage mainly about?

- A) Why libraries should require complete silence
- B) How materials and room arrangement help manage sound
- C) Why bookshelves are the best form of sound control
- D) How public buildings select durable flooring

36. How does the third paragraph develop the passage?

- A) It challenges the explanation given in the first paragraph.
- B) It describes the history of library construction.
- C) It shifts from sound-absorbing materials to the placement of spaces and activities.
- D) It compares libraries with several other public buildings.

37. Why does the author state that a library should not remove every sound?

- A) To suggest that acoustic materials are ineffective
- B) To criticize visitors who speak in study areas
- C) To show that hard surfaces are necessary everywhere
- D) To emphasize that useful communication and safety sounds must remain audible

38. Which conclusion is best supported by the passage?

- A) Effective sound control depends on matching design choices to how each area is used.
- B) A library with carpet does not need separate study spaces.
- C) Tight door seals are useful only in newly constructed buildings.
- D) Most unwanted library noise comes from alarms and conversations.

Literary Passage: The Unclaimed Apron

At the end of the community cooking workshop, Elena found a blue apron folded beneath a chair. A stitched silver fish marked one pocket. The other students had already left, and the building manager was stacking tables. Elena could have placed the apron in the lost-and-found box, but the fish seemed familiar. During the workshop, Mr. Ito had demonstrated how to shape dumplings. He had worn a plain white apron, yet he had mentioned that his first job was at his aunt's seafood café. Elena remembered seeing him carry two bags to his bicycle after class. She hurried outside, but the bicycle rack was empty. Instead of guessing, Elena returned to the kitchen and examined the sign-in sheet. Beside Mr. Ito's name was a note: "bringing supplies for Tuesday group." Elena called the program office and learned that he would return the next afternoon.

On Tuesday, Mr. Ito smiled when Elena held up the apron. It belonged not to him but to his aunt, who had sewn the fish years ago. He had brought it to show the class and had not noticed it missing. Before leaving, he asked Elena to help him pin the apron beside an old café photograph for the next workshop. Elena had set out to return a forgotten object; now she understood why it had been worth identifying its owner carefully.

39. What did Elena use to determine that Mr. Ito would return?

- A) A note beside his name on the sign-in sheet
- B) A label inside one of his supply bags
- C) A message attached to the bicycle rack
- D) A schedule posted by the building manager

40. Why does Elena avoid simply assuming that the apron belongs to Mr. Ito?

- A) She worries that he will not remember the workshop.
- B) She wants evidence before deciding who owns it.
- C) She believes the manager may have left it there.
- D) She plans to keep it if no one claims it.

41. Which statement best expresses a central idea of the passage?

- A) Cooking classes are most useful when families attend together.
- B) Valuable objects should always be displayed instead of used.
- C) Patient investigation can reveal the meaning connected to an object.
- D) Written records are more dependable than personal memories.

42. As used in the final sentence, "identifying" most nearly means

- A) describing the appearance of.
- B) making a public announcement about.
- C) attaching a permanent label to.
- D) determining the identity of.

Scientific Passage: A Cool Roof Above the Sidewalk

On a sunny day, a dark roof can absorb much of the sunlight that reaches it. The roof then releases some of that energy as heat, warming both the building and the nearby air. To reduce this effect, some buildings use “cool roofs,” which reflect more sunlight and release absorbed heat efficiently.

A cool roof is not necessarily bright white. Manufacturers can create lighter surfaces and specially coated materials in several colors. Researchers evaluate these products with instruments rather than judging by appearance alone.

One measurement records solar reflectance, the fraction of incoming sunlight a surface sends away. Another records thermal emittance, the surface’s ability to release absorbed heat.

The benefits depend on location and building design. In a hot climate, a cooler roof may reduce the need for air-conditioning. A building shaded by tall trees, however, receives less direct sunlight, so changing its roof may have a smaller effect. Insulation, roof angle, local weather, and energy prices also influence the result.

City planners sometimes study groups of roofs because many hot surfaces close together can raise neighborhood temperatures. Cool roofs alone cannot solve urban heat, but they can work alongside shade trees, parks, and light-colored pavement. The most effective plan usually combines several methods selected for local conditions.

43. According to the passage, what does solar reflectance measure?

- A) The fraction of incoming sunlight a surface sends away
- B) The amount of rainwater a roof can collect
- C) The rate at which insulation loses thickness
- D) The angle at which sunlight reaches a building

44. Why does the author mention a building shaded by tall trees?

- A) To argue that trees damage specially coated roofs
- B) To show that a roof’s benefit varies with local conditions
- C) To explain why shaded buildings require brighter roof colors
- D) To prove that roof angle is unimportant

45. How is the passage mainly organized?

- A) It lists roof types in order of cost and then recommends the cheapest.
- B) It presents opposing opinions before declaring one side correct.
- C) It explains a technology, describes how it is measured, and discusses when it is useful.
- D) It narrates the invention of a product from its first trial to modern use.

46. As used in paragraph 2, “fraction” most nearly means

- A) a disagreement.
- B) a broken piece.
- C) a mathematical rule.
- D) a portion of a whole.

Biographical-Style Passage: Maps Made from Questions

When surveyor Amara Bell arrived in the growing river town of Northbank in 1912, officials gave her a straightforward assignment: prepare a dependable street map. Existing maps showed major roads, but several lanes had changed names, and new workshops crowded the waterfront.

Bell began with instruments, measuring distances and recording angles. She soon discovered, however, that measurements could not explain every route. A narrow passage appeared to end at a fence, yet workers told her that a gate opened during market hours. A road marked "Mill Lane" was called "Dock Road" by delivery drivers. Bell therefore added interviews to her fieldwork. She asked residents how streets were used, when gates opened, and which landmarks helped visitors.

Her completed map included official names, but a separate guide recorded common local names and access notes. Some officials initially objected that these details made the project less tidy. Bell replied that a useful map should describe movement, not merely lines. After emergency crews used her access notes to reach a waterfront fire quickly, the guide became part of the town's standard records.

Bell's work illustrates a lasting principle of mapmaking: accuracy includes human experience. Measurements establish location, while observation and local knowledge reveal how places function. Neither source is sufficient by itself.

47. What information did Bell place in the separate guide?

- A) Construction costs for new waterfront roads
- B) Common street names and notes about access
- C) Biographies of officials who approved the map
- D) Predictions about future workshop locations

48. What can reasonably be inferred about the emergency crews?

- A) They had previously refused to use Bell's map.
- B) They changed the official name of Mill Lane.
- C) They benefited from information not shown by measurements alone.
- D) They asked residents to redraw the waterfront.

49. What is the author's main purpose in presenting Bell's work?

- A) To criticize officials for allowing towns to grow too quickly
- B) To compare early surveying instruments with modern ones
- C) To argue that informal street names should replace official names
- D) To demonstrate how local knowledge can strengthen technical accuracy

50. Which statement best summarizes the passage?

- A) Bell produced a more useful map by combining measurements with knowledge of how people used the town.
- B) Bell abandoned surveying when residents questioned the town's official street names.
- C) Northbank officials hired Bell mainly to create new routes for emergency crews.
- D) Northbank's waterfront expanded because Bell publicized its workshops.

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: A Hallway History Display

[1] Our class decided to create a hallway display about the history of the school building. [2] We first searched old yearbooks and interviewed two former students. [3] The building opened in 1958. [4] Next, small teams selected photographs that showed how classrooms, clubs, and school events had changed. [5] Because some photographs lacked dates, we compared clothing, furniture, and handwritten notes to place them in the correct decade. [6] We arranged the final display as a timeline, with short captions beneath each image. [7] Students passing through the hallway now often stop to examine details that once sat unnoticed in storage boxes. [8] The project taught us that local history can become vivid when evidence is carefully organized.

51. Which sentence should be deleted because it interrupts the logical development of the passage?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 5
- D) Sentence 7

52. Which revision of “place them in the correct decade” in sentence 5 is most precise?

- A) put them somewhere in time
- B) make a good guess about them
- C) estimate when each photograph was taken
- D) discover what the pictures were doing

53. Which transition would best begin sentence 6?

- A) On the contrary,
- B) Finally,
- C) For example,
- D) Meanwhile,

54. Which is the best way to combine sentences 1 and 2?

- A) Our class decided to create a hallway display, the history of the school building was searched in old yearbooks and interviews.
- B) Deciding to create a hallway display about the school building, two former students were interviewed by our class.
- C) Our class created a hallway history display, searching old yearbooks, and two former students.
- D) After deciding to create a hallway display about the school building’s history, our class searched old yearbooks and interviewed two former students.

Revision Passage: Starting a Seed Exchange

[1] A seed exchange allows gardeners to share extra seeds instead of discarding them. [2] Our community center could begin one with a cabinet, labeled envelopes, and a simple record sheet. [3] Participants would write the plant name and the year the seeds were collected on each envelope. [4] Clear labels matter, they help gardeners choose seeds suited to their space. [5] A volunteer could check the cabinet weekly and remove damaged packets. [6] The exchange should also provide brief instructions about proper seed storage. [7] These steps would keep the collection orderly and useful. [8] With modest supplies and regular care, the project could connect experienced gardeners with curious beginners.

55. Which revision best corrects sentence 4?

- A) Clear labels matter helping gardeners choose seeds suited to their space.
- B) Clear labels matter because they help gardeners choose seeds suited to their space.
- C) Clear labels matter, and because gardeners choose seeds suited to their space.
- D) Clear labels matter; because helping gardeners choose suitable seeds.

56. Where should the following sentence be placed? For example, seeds exposed to moisture may develop mold.

- A) After sentence 6
- B) After sentence 2
- C) After sentence 4
- D) After sentence 8

57. Which revision of sentence 5 is most concise without changing its meaning?

- A) A volunteer, being a person who helps, could make a weekly check of the cabinet and remove packets that are damaged.
- B) Every week, the cabinet could be checked by a volunteer who could remove any and all packets with damage.
- C) A volunteer could check on the cabinet in a weekly manner and take out the damaged seed packets that are there.
- D) A volunteer could inspect the cabinet weekly and remove damaged packets.

58. Which revision of sentence 8 uses parallel structure?

- A) With modest supplies, caring regularly, and beginners who are curious, the project could connect gardeners.
- B) The project, with supplies that are modest and regular care, connecting experienced gardeners with curious beginners.
- C) Requiring only modest supplies and regular care, the project could connect experienced gardeners with curious beginners.
- D) The project could connect gardeners who are experienced, beginners with curiosity, and regular care.

59. Which sentence is written correctly?

- A) Neither the coach nor the players was expecting the field to be closed.
- B) Neither the coach nor the players were expecting the field to be closed.
- C) Neither the coach or the players were expecting the field to be closed.
- D) Neither the coach nor the players is expecting the field were closed.

60. Choose the correct word to complete the sentence.

The two finalists performed equally well, but Lena spoke _____ during the question period.

- A) most confident
- B) more confidently than any speaker
- C) more confidently
- D) confidentlier

61. Which sentence expresses the idea most clearly?

- A) After Jordan placed the glass on the tray, he carried the tray to the table.
- B) Jordan put it on the tray and carried it there.
- C) Placing the glass there, the table was reached by Jordan with the tray.
- D) After putting it down, Jordan carried this over to that location.

62. Which line contains an error in standard written US English?

Line 1: The museum guide greeted Elena and me at the entrance.

Line 2: She showed us a gallery that had recently reopened.

Line 3: We thanked the guide before leaving.

Line 4: Between you and I, the final exhibit was the most impressive.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

63. Which line contains an error in standard written US English?

Line 1: By the time the bus arrived, the rain had stopped.

Line 2: Several students folded their umbrellas before boarding.

Line 3: The driver reminded everyone to keep the aisle clear.

Line 4: Each of the wet jackets were placed on a hook.

- A) Line 1
- B) Line 4
- C) Line 3
- D) Line 2

Revision Passage: Testing the Cafeteria Line

[1] Our student council studied why the cafeteria line moved slowly on certain days. [2] Council members timed the line during four lunch periods and noted where students stopped. [3] The cafeteria walls were painted pale green several years ago. [4] The observations showed that many students waited until reaching the counter to read the menu. [5] We suggested placing a large menu near the entrance so students could decide earlier. [6] We also recommended moving napkins and utensils to a separate table after the payment station. [7] The manager tested both changes for one week, the average wait became nearly two minutes shorter. [8] The trial showed how small adjustments based on direct observation can improve a familiar routine.

64. Which sentence should be deleted because it does not support the passage's central idea?

- A) Sentence 2
- B) Sentence 5
- C) Sentence 3
- D) Sentence 8

65. Which revision of sentence 2 is most precise?

- A) Council members timed how long students spent in line during four lunch periods and recorded the locations of repeated delays.
- B) Council members did some timing at lunch and wrote down things about the line.
- C) During lunch, the council watched everybody and made several useful notes.
- D) Four times, council members studied a line that was sometimes moving slowly.

66. Which revision best corrects sentence 7?

- A) The manager tested both changes for one week, with the average wait nearly two minutes shorter.
- B) The manager tested both changes. For one week and the average wait became nearly two minutes shorter.
- C) Testing both changes for one week, the average wait and the manager became shorter.
- D) The manager tested both changes for one week, and the average wait became nearly two minutes shorter.

67. Where would the following sentence best be placed?

This pattern suggested that providing information sooner might reduce the delay.

- A) Before sentence 1
- B) After sentence 4
- C) After sentence 6
- D) After sentence 8

Revision Passage: Building a Shadow Clock

[1] For a science project, my partner and I built a simple shadow clock in the school courtyard. [2] First, we placed a straight wooden stick upright in a bucket filled with packed sand. [3] The bucket was extremely heavy in weight because the sand inside it weighed a lot. [4] Beginning at nine in the morning, we marked the tip of the stick's shadow every hour. [5] Each mark included the time and the weather conditions. [6] Clouds sometimes made the shadow faint, but it became visible again when sunlight returned. [7] By midafternoon, the marks formed a curved path around the bucket. [8] Our observations demonstrated both the usefulness and the limits of measuring time with sunlight.

68. Which revision of sentence 3 is clearest and most concise?

- A) The bucket, having sand inside of it, was something that had a great deal of heaviness.
- B) Due to the fact that the sand weighed a lot, the bucket was heavy in weight.
- C) The packed sand made the bucket stable.
- D) The bucket was extremely heavy and not light because of its heavy sand.

69. Which revision best combines sentences 5 and 6?

- A) Each mark included the time and weather conditions, clouds sometimes made the shadow faint.
- B) Each mark included the time and weather conditions; although clouds sometimes made the shadow faint.
- C) Including the time and weather conditions, the clouds made each mark faint until sunlight.
- D) Although clouds sometimes made the shadow faint, each mark recorded the time and weather conditions whenever the shadow was visible.

70. Which additional detail would best support the conclusion in sentence 8?

- A) On a fully cloudy morning, we could not make a reliable time mark.
- B) The courtyard also contains two benches and a drinking fountain.
- C) My partner carried the pencil while I carried the notebook.
- D) Several students prefer digital watches with colorful bands.

71. Which revision of sentence 4 uses verb forms correctly?

- A) Beginning at nine in the morning, we had marked the shadow's tip every hour and stopping later.
- B) We began at nine in the morning, marking the tip of the stick's shadow every hour.
- C) We begin at nine in the morning and had marked the shadow's tip every hour.
- D) Beginning at nine in the morning, the shadow's tip was marking by us every hour.

72. Which sentence is written correctly?

- A) The committee have submitted its recommendation.
- B) The committee has submitted their recommendation.
- C) The committee has submitted its recommendation.
- D) The committee have submitted their recommendation.

73. Which sentence uses punctuation correctly?

- A) After the final bell rang we packed our materials, and left.
- B) After the final bell rang we packed our materials and left.
- C) After the final bell, rang we packed our materials and left.
- D) After the final bell rang, we packed our materials and left.

**74. Line 1: The museum guide described the first exhibit.
Line 2: She then led the visitors through two galleries.
Line 3: The final gallery contained fewer paintings then the first.
Line 4: Each visitor returned the map at the exit.**

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

**75. Line 1: The two puppies slept quiet beneath the table.
Line 2: Their owner filled a bowl with fresh water.
Line 3: Later, she divided the treats between the two puppies.
Line 4: Both animals woke when they heard the bag open.**

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

**76. Line 1: By the time the doors opened, Marcus had waited for twenty minutes.
Line 2: He checks his ticket once more before entering the theater.
Line 3: An usher directed him toward the balcony.
Line 4: His friends were already seated in the third row.**

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Recording Neighborhood Sounds

[1] Our media club created an audio portrait of the neighborhood surrounding the school. [2] Before recording, we listed sounds that might reveal how the area changes during the day. [3] At sunrise, one team captured delivery carts rolling over the sidewalk and birds calling from the park. [4] Another team recorded the afternoon, buses stopped, shop doors opened, and basketballs struck the playground pavement. [5] We labeled every clip with its location and time. [6] This careful catalog helped us arrange the sounds from early morning through evening. [7] The finished recording lasts eight minutes. [8] Listeners hear the neighborhood awaken, grow busy, and settle again without a narrator explaining the sequence.

77. Which revision best corrects sentence 4?

- A) Another team recorded the afternoon: buses stopped, shop doors opened, and basketballs struck the playground pavement.
- B) Another team recorded the afternoon buses, stopped shop doors, and basketballs striking the playground pavement.
- C) Another team recorded the afternoon, and buses stopped, shop doors opened, basketballs struck the playground pavement.
- D) Recording the afternoon, buses stopped, shop doors opened, and basketballs struck the playground pavement.

78. Which sentence would best follow sentence 5?

- A) Many students in the club also enjoy watching films.
- B) We also removed clips overwhelmed by wind or unrelated conversation.
- C) The school day begins before most neighborhood shops open.
- D) Audio files can require substantial computer storage.

79. Which revision of sentence 7 most effectively connects it to the passage's main idea?

- A) The finished recording is something that lasts for eight minutes altogether.
- B) Our final project has a length that is not especially long.
- C) In just eight minutes, the finished recording traces an entire day through its changing sounds.
- D) The recording, which is finished, contains sounds and lasts eight minutes.

80. Which revision best combines sentences 5 and 6 while showing cause and effect?

- A) We labeled every clip, and its location and time were in a careful catalog.
- B) Labeling clips with locations and times, the arrangement went from morning through evening.
- C) We arranged the sounds from morning through evening, but every clip had a label.
- D) Because we labeled every clip by location and time, we could arrange the sounds from early morning through evening.

Revision Passage: Repairing a Trail Sign

[1] A wooden sign beside the nature trail had become difficult to read after years of sun and rain. [2] Our service group photographed it before carefully removing the loose paint. [3] We consulted an older park map to confirm the wording and the direction of each arrow. [4] The sign was painted bright blue by the group on Saturday. [5] After the paint dried, we applied a clear outdoor coating for protection. [6] Finally, a park employee inspected the sign and helped us return it to its original post. [7] The restored sign now gives hikers clear information while preserving a familiar feature of the trail.

81. Which revision of sentence 4 is most direct and consistent with the passage?

- A) On Saturday, our group painted the sign bright blue.
- B) Bright blue was the color that was painted onto the sign by our group.
- C) The sign, it was painted by the group in a bright blue color on Saturday.
- D) On Saturday, there was bright blue paint applied to the sign by us.

82. Which detail best supports the group's decision to consult the older park map in sentence 3?

- A) The map was printed on thick, cream-colored paper.
- B) Some letters and one arrow on the sign had faded completely.
- C) The park employee stored several maps in an office drawer.
- D) Group members disagreed about which paintbrush to use.

83. Which revision of sentence 5 most precisely explains the purpose of the coating?

- A) After the paint dried, we put something clear over it.
- B) We applied a clear coating, which was a good thing to do.
- C) After the paint dried, we applied a clear coating to shield the sign from moisture and sunlight.
- D) The paint dried before a clear outdoor coating was applied for various reasons.

84. Which sentence uses pronouns correctly?

- A) Me and Elena checked the equipment before practice.
- B) The coach gave Marcus and I the new schedule.
- C) Between you and I, the shortcut seems confusing.
- D) The final decision belongs to Carla and him.

85. Which sentence is written correctly?

- A) By the time the guests arrived, we had already arranged the chairs.
- B) By the time the guests arrived, we already arrange the chairs.
- C) By the time the guests arrive, we had already arranged the chairs yesterday.
- D) By the time the guests had arrived, we have already arranged the chairs.

86. Which sentence places the modifying phrase clearly?

- A) Covered with frost, Mia photographed the leaves beside the path.
- B) Mia photographed the leaves covered with frost beside the path.
- C) Mia, beside the path covered with frost, photographed the leaves.
- D) Covered with frost beside the path, the leaves were photographed by Mia quickly.

87. Which line contains an error in standard written US English?

Line 1: The committee reviewed each proposal carefully.

Line 2: Neither of the two plans includes a covered entrance.

Line 3: Several members requests additional cost estimates.

Line 4: A final recommendation will be announced tomorrow.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

88. Which line contains an error in standard written US English?

Line 1: Rosa opened the package at the kitchen table.

Line 2: Inside were three jars wrapped in paper.

Line 3: She placed each jar carefully on the shelf.

Line 4: Then she recycle the box and packing material.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Designing a Portable Book Display

[1] Our reading club wanted a display that could travel between classrooms. [2] We first listed the display's requirements: it had to hold twelve books, fit through doorways, and be easy for two students to carry. [3] One student owns a red backpack with three pockets. [4] We sketched several designs and built a small cardboard model of the strongest one. [5] During a test, the model tipped when all the books were placed on one side. [6] We widened the base and arranged the shelves so equal weight rested on each side. [7] After a second test, the display remained steady while being moved. [8] The finished wooden version now carries each month's book selections to four classrooms.

89. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

90. Which revision most effectively combines sentences 4 and 5?

- A) We sketched several designs, a small cardboard model was built, and it tipped during testing.
- B) After sketching several designs, we built a small cardboard model of the strongest one, but it tipped when books were placed on one side.
- C) Sketching several designs and building a small cardboard model, the one-sided books caused tipping.
- D) We built a model after sketches, tipping because the strongest one was cardboard.

91. Which revision of sentence 7 is most precise?

- A) The next test was better in a way that mattered.
- B) After that, it stayed good while we did things with it.
- C) In the second test, the display stayed balanced as two students carried it across the room.
- D) The display remained steady, which was nice, during another test.

92. Which sentence would provide the most effective conclusion after sentence 8?

- A) Wood can be purchased in boards of several different lengths.
- B) Other clubs meet in the classrooms on certain afternoons.
- C) Our first cardboard model was smaller than the final display.
- D) By testing and revising our design, we created a useful display that meets the club's original needs.

Revision Passage: Interviewing a Local Baker

[1] For the school podcast, I interviewed Ms. Alvarez, who begins baking before sunrise. [2] I prepared questions about her schedule, training, and favorite tools. [3] At first, my questions followed no clear order, so the conversation moved abruptly between topics. [4] I reorganized them into groups and added short follow-up questions. [5] During the interview, Ms. Alvarez described learning to judge dough by touch rather than by a timer alone. [6] Her explanation was extremely fascinating and very interesting to me. [7] When I edited the recording, I kept the pauses that allowed listeners to consider her advice. [8] The final episode showed not only what a baker does but also how experience guides quick decisions.

93. Which revision of sentence 4 best clarifies the relationship between the two actions?

- A) After reorganizing the questions into related groups, I added short follow-up questions to each group.
- B) I reorganized questions into groups, adding was done with short follow-up questions.
- C) The questions were groups, and short follow-up questions, I added them.
- D) Reorganizing into groups, the follow-up questions were short.

94. Which revision of sentence 6 is most concise and effective?

- A) Her explanation was fascinating and interesting in an extreme way to me.
- B) I found her explanation fascinating.
- C) Her explanation was very extremely interesting and fascinating.
- D) To me personally, I myself found the explanation interesting.

95. Which revision of sentence 1 uses punctuation correctly?

- A) For the school podcast I interviewed, Ms. Alvarez who begins baking before sunrise.
- B) For the school podcast, I interviewed Ms. Alvarez who, begins baking before sunrise.
- C) For the school podcast, I interviewed Ms. Alvarez, who begins baking before sunrise.
- D) For the school podcast I interviewed Ms. Alvarez who begins, baking before sunrise.

96. Which sentence is written correctly?

- A) One of the musicians practices in the auditorium each morning.
- B) The box of old photographs were stored upstairs.
- C) Neither of the two routes lead directly to the station.
- D) Every one of the lanterns have a paper label.

97. Which sentence uses the pronoun correctly?

- A) Please give the permission slips to Marcus or I.
- B) The science teacher spoke with Lena and me after class.
- C) Her and Talia prepared the equipment together.
- D) Between you and she, the shortcut is difficult to follow.

98. Which numbered line contains an error?

Line 1: The volunteers had completed the mural by noon.

Line 2: They cleaned their brushes before eating lunch.

Line 3: Later, the director thanks them for their careful work.

Line 4: Everyone returned home before the rain began.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

99. Which numbered line contains an error?

Line 1: The new telescope is easier to adjust than the old one.

Line 2: Its controls respond smoothly in cold weather.

Line 3: The instruction card is also clearly written.

Line 4: Of the two models, the new telescope is the easiest to transport.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

100. Which numbered line contains an error?

Line 1: Before the doors opened, each performer checked a costume.

Line 2: The stage manager reviewed the schedule with the cast.

Line 3: Several actors placed their props beside the curtain.

Line 4: The musicians tuned there instruments in the orchestra pit.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. A tank is $\frac{3}{5}$ full. After 18 liters are added, it is $\frac{3}{4}$ full. What is the tank's total capacity?

- A) 90 liters
- B) 120 liters
- C) 72 liters
- D) 150 liters

102. What is the value of $2.4 \div 0.06$?

- A) 0.4
- B) 4
- C) 400
- D) 40

103. A price is reduced by 20% and then increased by 20% of the reduced price. If the original price was \$75, what is the final price?

- A) \$72
- B) \$75
- C) \$70
- D) \$60

104. The first four terms of a sequence are 5, 9, 17, 33. Each term after the first is found by applying the same rule. What is the next term?

- A) 49
- B) 61
- C) 65
- D) 66

105. Which inequality is equivalent to $4x - 7 \leq 21$?

- A) $x \leq 3 \frac{1}{2}$
- B) $x \leq 7$
- C) $x \geq 7$
- D) $x \leq 28$

106. A delivery service charges a fixed fee plus \$3 per mile. A 6-mile delivery costs \$23. Which expression gives the cost, in dollars, of an m -mile delivery?

- A) $3m + 5$
- B) $5m + 3$
- C) $3(m + 23)$
- D) $23m - 6$

107. The mean of five quiz scores is 84. Four scores are 78, 81, 86, and 89. What is the fifth score?

- A) 82
- B) 84
- C) 88
- D) 86

108. A bag contains 4 red tiles, 3 blue tiles, and 2 green tiles. One tile is selected, not replaced, and then a second tile is selected. What is the probability that both tiles are green?

- A) $\frac{2}{9}$
- B) $\frac{1}{9}$
- C) $\frac{1}{36}$
- D) $\frac{1}{18}$

109. Two parallel lines are crossed by a transversal. One of the acute angles formed measures 67° . What is the measure of each obtuse angle formed?

- A) 113°
- B) 67°
- C) 23°
- D) 134°

110. A right triangle has legs of length 9 units and 12 units. What is its area?

- A) 42 square units
- B) 54 square units
- C) 108 square units
- D) 126 square units

111. A scale drawing uses 1 centimeter to represent 2.5 meters. A rectangular room measures 3.6 centimeters by 2.8 centimeters on the drawing. What is the actual perimeter of the room?

- A) 16 meters
- B) 25 meters
- C) 32 meters
- D) 40 meters

112. A train leaves at 10:48 a.m. and arrives at 1:17 p.m. It stops for 22 minutes during the trip. How long is the train moving?

- A) 2 hours 51 minutes
- B) 2 hours 29 minutes
- C) 1 hour 57 minutes
- D) 2 hours 7 minutes

113. A bag contains $2\frac{2}{3}$ pounds of rice. A cook uses $\frac{3}{4}$ pound. How much rice remains?

- A) $1\frac{7}{12}$ pounds
- B) $1\frac{5}{6}$ pounds
- C) $2\frac{1}{12}$ pounds
- D) $1\frac{11}{12}$ pounds

114. What is the least common multiple of 12, 18, and 30?

- A) 180
- B) 90
- C) 120
- D) 360

115. At a fair, 7 out of every 20 visitors bought a program. What percent of the visitors bought a program?

- A) 28%
- B) 30%
- C) 35%
- D) 40%

116. The rule for a function is $y = 2x^2 - 3$. What is y when $x = -4$?

- A) 19
- B) 29
- C) 35
- D) 61

117. Three consecutive integers have a sum of 72. What is the greatest of the three integers?

- A) 23
- B) 24
- C) 26
- D) 25

118. The numbers of pages Luis read on five days were 18, 24, 15, 27, and 21. What was the median number of pages?

- A) 18
- B) 24
- C) 21
- D) 22

119. A box contains 5 red cards, 3 blue cards, and 2 green cards. One card is selected, replaced, and then a second card is selected. What is the probability that both cards are blue?

- A) $\frac{9}{100}$
- B) $\frac{3}{10}$
- C) $\frac{3}{20}$
- D) $\frac{6}{100}$

120. A music club recorded attendance for four meetings: 18, 22, 25, and 19 students. How many students must attend the fifth meeting for the mean attendance to be 21?

- A) 19
- B) 20
- C) 22
- D) 21

121. Each interior angle of a regular polygon measures 135° . How many sides does the polygon have?

- A) 6
- B) 8
- C) 9
- D) 10

122. A square has a diagonal of length $10\sqrt{2}$ centimeters. What is the area of the square?

- A) 50 cm^2
- B) 20 cm^2
- C) 100 cm^2
- D) 200 cm^2

123. A rectangular prism is 8 centimeters long, 5 centimeters wide, and 3 centimeters high. What is its total surface area?

- A) 79 cm^2
- B) 94 cm^2
- C) 120 cm^2
- D) 158 cm^2

124. A container holds 3.5 liters of juice. How many 250-milliliter cups can be filled completely? Use 1 liter = 1,000 milliliters.

- A) 14
- B) 12
- C) 16
- D) 18

125. A cyclist travels 27 miles in 1 hour 48 minutes at a constant rate. What is the cyclist's average speed in miles per hour?

- A) 12
- B) 15
- C) 18
- D) 24

126. What is the value of $\frac{3}{5} \div \frac{9}{10}$?

- A) $\frac{2}{3}$
- B) $\frac{3}{10}$
- C) $\frac{5}{6}$
- D) $\frac{3}{2}$

127. A jacket's price is reduced from \$64 to \$48. What percent decrease is this?

- A) 20%
- B) 25%
- C) 30%
- D) $33\frac{1}{3}\%$

128. Which number is both a factor of 84 and a multiple of 7?

- A) 12
- B) 18
- C) 21
- D) 28

129. The first four terms of a sequence are 5, 11, 23, and 47. Which rule generates each term after the first?

- A) Add 6.
- B) Add consecutive even numbers.
- C) Multiply by 3, then subtract 4.
- D) Multiply by 2, then add 1.

130. A taxi fare is modeled by $F = 3.50 + 2m$, where m is the number of miles traveled. What is the fare for 7 miles?

- A) \$17.50
- B) \$14.00
- C) \$18.50
- D) \$24.50

131. Three consecutive integers have a sum of 72. What is the greatest integer?

- A) 23
- B) 25
- C) 24
- D) 26

132. The numbers of pages read by five students are 18, 22, 22, 27, and 31. If a sixth student's result is included, the mean becomes 25 pages. How many pages did the sixth student read?

- A) 25
- B) 28
- C) 30
- D) 22

133. A spinner has eight equal sections numbered 1 through 8. What is the probability of landing on a number that is prime or composite?

- A) $\frac{1}{2}$
- B) $\frac{3}{4}$
- C) $\frac{5}{8}$
- D) $\frac{7}{8}$

134. An isosceles triangle has a vertex angle of 38° . What is the measure of each base angle?

- A) 71°
- B) 76°
- C) 104°
- D) 142°

135. A circle has a diameter of 14 centimeters. Using $\pi = 22/7$, what is its circumference?

- A) 22 cm
- B) 44 cm
- C) 49 cm
- D) 154 cm

136. A scale drawing uses 1 centimeter to represent 2.5 meters. A wall measures 7.2 centimeters in the drawing. What is the actual length of the wall?

- A) 9.7 m
- B) 14.4 m
- C) 18 m
- D) 25 m

137. A tank contains 3.6 liters of water. How many 150-milliliter cups can be completely filled? Use 1 liter = 1,000 milliliters.

- A) 18
- B) 20
- C) 22
- D) 24

138. A recipe uses $2\frac{1}{4}$ cups of oats for 3 batches. At the same rate, how many cups are needed for 8 batches?

- A) 6 cups
- B) $5\frac{1}{4}$ cups
- C) $6\frac{3}{4}$ cups
- D) $7\frac{1}{2}$ cups

139. What is the value of $18 - 3 \times (4 + 2) \div 2$?

- A) 0
- B) 6
- C) 9
- D) 15

140. A number is increased by 15% and becomes 92. What was the original number?

- A) 75
- B) 80
- C) 82
- D) 107

141. The perimeter of a rectangle is 46 centimeters. Its length is 5 centimeters greater than its width. What is the width?

- A) 8 cm
- B) 10 cm
- C) 12 cm
- D) 9 cm

142. For the function $f(x) = 4x - 7$, which value of x makes $f(x) = 21$?

- A) 5
- B) 6
- C) 7
- D) 8

143. The numbers of pages read by five students are 18, 24, 24, 29, and 35. What is the range?

- A) 17
- B) 11
- C) 24
- D) 26

144. A spinner has 8 equal sections: 3 blue, 2 green, 2 yellow, and 1 red. What is the probability of landing on a section that is neither blue nor red?

- A) $\frac{1}{4}$
- B) $\frac{3}{8}$
- C) $\frac{5}{8}$
- D) $\frac{1}{2}$

145. A bus carried 24 passengers on Monday, 30 on Tuesday, 27 on Wednesday, and 35 on Thursday. How many passengers must it carry Friday for the five-day mean to be 30?

- A) 30
- B) 34
- C) 36
- D) 38

146. Two similar triangles have corresponding side lengths 6 and 15. If another side of the smaller triangle is 8, what is the corresponding side of the larger triangle?

- A) 12
- B) 17
- C) 20
- D) 24

147. A circle has a circumference of 31.4 centimeters. Using $\pi = 3.14$, what is its radius?

- A) 5 cm
- B) 10 cm
- C) 15.7 cm
- D) 20 cm

148. A rectangular prism has a volume of 180 cubic inches. Its length is 9 inches and its width is 5 inches. What is its height?

- A) 2 inches
- B) 3 inches
- C) 5 inches
- D) 4 inches

149. A 14-foot board is cut into pieces that are each 1 foot 9 inches long. How many complete pieces can be cut?

- A) 6
- B) 8
- C) 9
- D) 10

150. A scale drawing uses 3 centimeters to represent 12 meters. A hallway measures 8.5 centimeters on the drawing. What is its actual length?

- A) 28 meters
- B) 30 meters
- C) 34 meters
- D) 36 meters

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Complete the 2×2 matrix.

Top-left: ○

Top-right: ○ ○

Bottom-left: ●

Bottom-right: ?

Across each row, one identical shape is added. Down each column, open shapes become filled.

- A) ○ ●
- B) ■ ■
- C) ● ●
- D) ● ● ●

152. Complete the 2×2 matrix.

Top-left: ↑

Top-right: →

Bottom-left: ↓

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise. Down each column, it rotates 180°.

- A) ←
- B) ↑
- C) →
- D) ↓

153. Complete the 2×2 matrix.

Top-left: Left ○ | Right □

Top-right: Left □ | Right ○

Bottom-left: Left ● | Right ■

Bottom-right: ?

Across each row, the two shapes exchange positions. Down each column, open shapes become filled.

- A) Left ● | Right ■
- B) Left □ | Right ○
- C) Left ○ | Right □
- D) Left ■ | Right ●

154. Complete the 3×3 matrix.

Row 1: ○ | □ | ○ □

Row 2: △ | ◇ | △ ◇

Row 3: ● | ■ | ?

In each row, the third cell combines the shapes in the first two cells in the same left-to-right order.

- A) ○ ■
- B) ● ■
- C) ■ ●
- D) ● □

155. Complete the 3×3 matrix.

Row 1: ○ | ● | ○

Row 2: □ | ■ | □

Row 3: △ | ▲ | ?

In each row, the same shape alternates open, filled, open.

- A) ▲
- B) ◇
- C) △
- D) ■

156. A square sheet is folded in half from left to right. One hole is punched near the upper-right corner of the folded rectangle. Where will the holes appear when the sheet is unfolded?

- A) Near both upper corners
- B) Near both left corners
- C) Near the upper-left and lower-right corners
- D) Near all four corners

157. A rectangular sheet is folded in half from bottom to top, then in half from right to left. One hole is punched at the center of the final folded rectangle. How many holes will appear when the sheet is completely unfolded?

- A) 2
- B) 4
- C) 6
- D) 8

158. A square sheet is folded along the diagonal from the upper-left corner to the lower-right corner. A hole is punched near the upper-right corner of the folded triangle, away from the fold. What appears after unfolding?

- A) One hole on the diagonal
- B) Four holes, one near each corner
- C) Two holes near the upper-left and lower-right corners
- D) Two holes reflected across the diagonal, near the upper-right and lower-left corners

159. A square sheet is folded in half from top to bottom. A semicircular cut is made along the middle of the horizontal fold edge. What appears when the sheet is unfolded?

- A) One complete circular hole centered on the horizontal centerline
- B) Two separate semicircular cuts at the top and bottom edges
- C) One semicircular cut at the left edge
- D) Two complete circular holes beside the centerline

160. Figures in the group:

1. ○ ● ○

2. □ ■ □

3. △ ▲ △

Which choice belongs in the same group?

- A) ○ □ ○
- B) ◆ ◇ ◆
- C) ◇ ◆ ◇
- D) ★ ★ ☆

161. Figures in the group:

1. ↑ → ↓

2. → ↓ ←

3. ↓ ← ↑

Which choice belongs in the same group?

A) ↑ ↓ →

B) ← ↓ →

C) → ↑ ←

D) ← ↑ →

162. Figures in the group:

1. Outer ○ | Center ■ | Outer ○

2. Outer □ | Center ▲ | Outer □

3. Outer △ | Center ◆ | Outer △

Which choice belongs in the same group?

A) Outer ○ | Center □ | Outer ●

B) Outer ◇ | Center ★ | Outer ◇

C) Outer ■ | Center ○ | Outer ■

D) Outer ☆ | Center ☆ | Outer ★

163. In each row, the third cell is formed by removing from the first cell every symbol that also appears in the second cell. Order is preserved.

Row 1: ○ □ △ | □ | ○ △

Row 2: ★ ◇ + | ★ + | ?

Which cell completes the matrix?

A) ★ ◇

B) ◇

C) ◇ +

D) ★

164. Consider this 2 × 2 matrix:

Top-left: Arrow ↑ outside an open circle ○

Top-right: Arrow → inside an open circle ○

Bottom-left: Arrow ← outside a filled circle ●

Bottom-right: ?

Moving right, the arrow rotates 90° clockwise and moves inside the circle. Moving down, the arrow rotates 90° counterclockwise and the circle becomes filled.

Which figure completes the matrix?

A) Arrow ↑ outside a filled circle ●

B) Arrow ↓ outside a filled circle ●

C) Arrow ← inside an open circle ○

D) Arrow ↑ inside a filled circle ●

165. Consider this 3 × 3 matrix:

Row 1: ○ | □ | △

Row 2: □ | △ | ○

Row 3: △ | ○ | ?

Every row and column contains each shape exactly once.

A) □

B) ○

C) △

D) ◇

166. In each row, the second cell is the horizontal reflection of the first. The third cell changes both reflected shapes from open to filled.

Row 1: Left ○ | Right □ ; Left □ | Right ○ ; Left ■ | Right ●

Row 2: Left △ | Right ◇ ; Left ◇ | Right △ ; ?

Which cell completes the matrix?

- A) Left ▲ | Right ◆
- B) Left △ | Right ◇
- C) Left ◆ | Right ▲
- D) Left ◇ | Right △

167. A square sheet is folded from left to right along the vertical centerline. The folded rectangle is then folded from bottom to top along its horizontal centerline. One hole is punched near the upper-right corner of the final folded square. What pattern appears when the sheet is completely unfolded?

- A) Four holes, one near each corner
- B) Two holes near the two upper corners
- C) Four holes centered along the four edges
- D) Two holes near the upper-right and lower-left corners

168. A rectangular sheet is folded in half from top to bottom. A small triangular cut is made in the middle of the folded sheet's left outer edge, not on the fold. What appears after the sheet is unfolded?

- A) One diamond opening at the center
- B) Four triangular openings near the corners
- C) Two matching triangular notches on the left edge, mirrored across the horizontal centerline
- D) Two notches, one on the left edge and one on the right edge

169. A square sheet is folded along the diagonal from the upper-left corner to the lower-right corner, forming a triangular packet. The packet is folded in half by bringing its upper-left corner to its lower-right corner. One hole is punched away from all folded edges. How many holes appear when the sheet is completely unfolded?

- A) 2
- B) 3
- C) 6
- D) 4

170. A rectangular sheet is folded from right to left along the vertical centerline. A semicircular cut is made at the midpoint of the folded sheet's top outer edge, away from the vertical fold. What appears when the sheet is unfolded?

- A) One circular hole at the center
- B) Two matching semicircular notches along the top edge
- C) Two semicircular notches along the left edge
- D) Four semicircular notches, one on each edge

171. The following arrangements belong to the same group:

○ above ●

□ above ■

△ above ▲

Which arrangement belongs to the same group?

- A) ◆ above ◇
- B) ★ above ☆
- C) ◇ above ◆
- D) ○ above ■

172. The following arrangements belong to the same group:

↑ ○ ↓
→ □ ←
↓ △ ↑

Which arrangement belongs to the same group?

- A) ← ◇ ↑
- B) ← ◇ →
- C) ↑ ◆ →
- D) → ○ ↓

173. The following figures belong to the same group:

Open circle ○ containing +

Open square □ containing +

Open diamond ◇ containing +

Which figure belongs to the same group?

- A) Open triangle △ containing +
- B) Filled triangle ▲ containing +
- C) Open triangle △ containing ×
- D) + containing an open triangle △

174. The following arrangements belong to the same group:

○ □ ○
△ ◇ △
☆ + ☆

Which arrangement belongs to the same group?

- A) ○ ○ □
- B) ◆ □ ◆
- C) + ☆ +
- D) □ △ □

175. Each row follows the same rule.

Row 1: ○ | ● | ○●

Row 2: □ | ■ | □■

Row 3: △ | ▲ | ?

Which cell completes the matrix?

- A) ▲△
- B) △▲
- C) △△
- D) ▲▲

176. In each row, the arrow in the second cell is the first arrow rotated 90° clockwise, and the third is rotated another 90° clockwise.

Row 1: ↑ | → | ↓

Row 2: ← | ↑ | →

Row 3: ↓ | ← | ?

Which arrow completes the matrix?

- A) ↑
- B) →
- C) ↓
- D) ←

177. In each row, the third cell contains the shape appearing in both of the first two cells.

Row 1: ○□ | □△ | □

Row 2: △◇ | ○△ | △

Row 3: ☆○ | ◇☆ | ?

Which cell completes the matrix?

- A) ○
- B) ◇
- C) ☆
- D) ○◇

178. The filled shape moves one corner clockwise in each cell.

Cell 1: Top-left ●; other corners ○

Cell 2: Top-right ●; other corners ○

Cell 3: Bottom-right ●; other corners ○

What is Cell 4?

- A) Top-left ●; other corners ○
- B) Top-right ●; other corners ○
- C) Bottom-right ●; other corners ○
- D) Bottom-left ●; other corners ○

179. Each row changes the first cell by exchanging open and filled forms while keeping the shapes and order unchanged.

Row 1: ○■ | ●□

Row 2: △◆ | ▲◇

Row 3: □★ | ?

Which cell completes the matrix?

- A) □☆
- B) ■☆
- C) ■★
- D) □★

180. A square sheet is folded from bottom to top along the horizontal centerline. It is then folded along the diagonal of the folded rectangle from its lower-left corner to its upper-right corner. One hole is punched away from every edge and fold. How many holes appear after complete unfolding?

- A) 4
- B) 2
- C) 6
- D) 8

181. A rectangular sheet is folded from left to right along the vertical centerline. Two holes are punched, one near the top outer edge and one near the bottom outer edge, at different distances from the vertical fold. What appears after unfolding?

- A) Two holes on the right half only
- B) Two holes aligned on the vertical centerline
- C) Four holes forming two left-right mirror pairs
- D) Eight holes forming four mirror pairs

182. A square sheet is folded along the diagonal from the upper-right corner to the lower-left corner. A triangular cut is made in the middle of the diagonal folded edge. What appears when the sheet is unfolded?

- A) One triangular notch on an outside edge
- B) Two separate triangular holes away from the diagonal
- C) Four triangular holes around the center
- D) One diamond-shaped hole centered on the diagonal

183. A square sheet is folded in half from right to left and then folded again from top to bottom. A small cut is made at the corner of the final folded square where the two folded edges meet. What appears after complete unfolding?

- A) Four matching corner cuts, one at each corner of the original sheet
- B) One cut at the center of the original sheet
- C) Two cuts along the top edge only
- D) Four cuts surrounding the center

184. Figures in the group:

- 1. ○ inside □
- 2. △ inside ◇
- 3. ☆ inside ○

In every figure, the inner shape has fewer straight sides than the outer shape. Which figure belongs to the same group?

- A) ○ inside △
- B) □ inside △
- C) ◇ inside ○
- D) ▲ inside ○

185. Arrangements in the group:

- 1. ○ ● ○
- 2. □ ■ □
- 3. △ ▲ △

Each arrangement has matching open shapes on the outside and the filled form of that shape in the center.

Which arrangement belongs to the same group?

- A) ◆ ◇ ◆
- B) ☆ ☆ ★
- C) ◇ ◆ ◇
- D) ● ○ ●

186. Figures in the group:

- 1. Top: ↑; Bottom: ↓
- 2. Left: ←; Right: →
- 3. Upper-left: ↖ is not used; instead, Top-left: ← and Bottom-right: →

For all three figures, the two arrows point away from the center along the line joining them. Which choice belongs to the same group?

- A) Top: ↓; Bottom: ↑
- B) Left: →; Right: ←
- C) Top: ↑; Bottom: ↑
- D) Left: ←; Right: →

187. Arrangements in the group:

1. ○ | □□ | △△△

2. ◇ | ○○ | ☆☆☆

3. □ | △△ | ◇◇◇

Each arrangement increases from one symbol to two matching symbols to three matching symbols, using a different shape in each position.

Which arrangement belongs to the same group?

A) ☆ | ☆☆☆ | ○○○

B) △ | □□ | ○○○

C) ○○ | □ | △△△

D) ◇ | ○○ | ◇◇◇

188. In each row, the third cell contains the symbols that appear in exactly one of the first two cells.

Row 1: ○ + | + □ | ○ □

Row 2: △ ☆ | ☆ ◆ | △ ◆

Row 3: ◇ ■ | ■ ● | ?

Which cell completes the matrix?

A) ■ ●

B) ◇ ■

C) ◇ ■ ●

D) ◇ ●

189. Each cell contains one arrow above one shape.

Top-left: ↑ above ○

Top-right: → above ●

Bottom-left: ← above ●

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise and the shape changes from open to filled or filled to open. Which completes the matrix?

A) ↑ above ○

B) ↓ above ○

C) ↓ above ●

D) → above ○

190. The position of a star moves one corner clockwise in each cell, while its form alternates open and filled.

Cell 1: top-left ☆

Cell 2: top-right ★

Cell 3: bottom-right ☆

Cell 4: ?

Which is Cell 4?

A) top-left ★

B) bottom-left ★

C) bottom-left ☆

D) top-right ☆

191. In each row, the third cell is formed by reversing the order of the first cell and then changing each shape from open to filled.

Row 1: ○ △ | △ ○ | ▲ ●

Row 2: □ ◇ | ◇ □ | ◆ ■

Row 3: ☆ ○ | ○ ☆ | ?

Which cell completes Row 3?

- A) ★ ●
- B) ● ☆
- C) ● ★
- D) ☆ ○

192. A square sheet is folded along the diagonal from upper-left to lower-right. A circular hole is punched near the upper-right corner of the folded triangle, away from the fold. What appears after the sheet is unfolded?

- A) One hole near the upper-right corner
- B) Four holes, one near each corner
- C) Two holes near the upper-left and lower-right corners
- D) Two holes reflected across the diagonal, near the upper-right and lower-left corners

193. A rectangular sheet is folded from left to right along its vertical centerline. It is then folded from bottom to top along its horizontal centerline. A small triangular cut is made at the folded packet's outer upper-right corner. What appears after unfolding?

- A) One triangular cut at the center
- B) Four matching corner cuts, one at each corner
- C) Two cuts along the right edge
- D) Four cuts along the centerlines

194. A square sheet is folded from top to bottom along the horizontal centerline. A semicircular notch is cut from the folded edge, left of center. What appears after unfolding?

- A) One complete circular hole left of center on the horizontal centerline
- B) Two semicircular notches on the top edge
- C) Two complete holes mirrored left and right
- D) One semicircular notch on the bottom edge

195. A rectangular sheet is folded from right to left along the vertical centerline. Two holes are punched in the folded sheet: one near the upper-left corner and one near the lower-left corner, both away from the fold. What appears after unfolding?

- A) Two holes along the left edge
- B) Four holes in one horizontal row
- C) Two holes near the upper corners only
- D) Four holes, forming upper and lower pairs reflected across the vertical centerline

196. These arrangements share one rule:

○ | ●●

□ | ■■

△ | ▲▲

Which arrangement follows the same rule?

- A) ◇ | ◇◇
- B) ◇ | ◆◆
- C) ◆ | ◇◇
- D) ◇◇ | ◆

197. These arrangements share one rule:

→ ○ ←
→ □ ←
→ △ ←

Which arrangement follows the same rule?

- A) ← ◇ →
- B) ← ◆ ←
- C) → ◇ ←
- D) ↑ ◇ →

198. These figures share one rule:

Outer □ containing inner ○

Outer △ containing inner □

Outer ◇ containing inner △

The inner shape of each figure is the outer shape of the preceding figure. Which figure can come next?

- A) Outer ○ containing inner ◇
- B) Outer △ containing inner ◇
- C) Outer ○ containing inner ○
- D) Outer ☆ containing inner ◇

199. These arrangements share one rule:

Top: ○ | Center: + | Bottom: ○

Top: □ | Center: × | Bottom: □

Top: △ | Center: - | Bottom: △

Which arrangement follows the same rule?

- A) Top: ◇ | Center: ◇ | Bottom: +
- B) Top: ◆ | Center: + | Bottom: ◇
- C) Top: ◇ | Center: | | Bottom: ◇
- D) Top: + | Center: ◇ | Bottom: +

200. These sequences share one rule:

○ ● ○ ●

□ ■ □ ■

△ ▲ △ ▲

Which sequence follows the same rule?

- A) ◇ ◆ ◇ ◆
- B) ◇ ◇ ◆ ◆
- C) ◆ ◇ ◆ ◇
- D) ◇ ◆ ◆ ◇

Answer Key

Question numbers continue across all five sections.

Reading

1. B	2. C	3. A	4. D	5. A	6. D	7. B	8. C
9. D	10. A	11. C	12. B	13. A	14. A	15. B	16. C
17. D	18. D	19. B	20. A	21. C	22. C	23. A	24. B
25. D	26. A	27. B	28. C	29. D	30. A	31. B	32. C
33. D	34. A	35. B	36. C	37. D	38. A	39. A	40. B
41. C	42. D	43. A	44. B	45. C	46. D	47. B	48. C
49. D	50. A						

Written Expression

51. A	52. C	53. B	54. D	55. B	56. A	57. D	58. C
59. B	60. C	61. A	62. D	63. B	64. C	65. A	66. D
67. B	68. C	69. D	70. A	71. B	72. C	73. D	74. C
75. A	76. B	77. A	78. B	79. C	80. D	81. A	82. B
83. C	84. D	85. A	86. B	87. C	88. D	89. A	90. B
91. C	92. D	93. A	94. B	95. C	96. A	97. B	98. C
99. D	100. D						

Mathematics

101. B	102. D	103. A	104. C	105. B	106. A	107. D	108. C
109. A	110. B	111. C	112. D	113. D	114. A	115. C	116. B
117. D	118. C	119. A	120. D	121. B	122. C	123. D	124. A
125. B	126. A	127. B	128. C	129. D	130. A	131. B	132. C
133. D	134. A	135. B	136. C	137. D	138. A	139. C	140. B
141. D	142. C	143. A	144. D	145. B	146. C	147. A	148. D
149. B	150. C						

Ability

151. C	152. A	153. D	154. B	155. C	156. A	157. B	158. D
159. A	160. C	161. D	162. B	163. B	164. D	165. A	166. C
167. A	168. C	169. D	170. B	171. C	172. B	173. A	174. D
175. B	176. A	177. C	178. D	179. B	180. A	181. C	182. D
183. B	184. A	185. C	186. D	187. B	188. D	189. A	190. B
191. C	192. D	193. B	194. A	195. D	196. B	197. C	198. D

199. C	200. A						
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Detailed Explanations

Reading

1. TACHS-R2-001 - Answer B: A bent wire pressed against its side.
Maya discovers that the wire has bent against the bell's side, preventing a clear sound.
2. TACHS-R2-002 - Answer C: The sound connects him with memories of his first shop.
Grandfather's repeated ringing after the repair suggests that the familiar sound carries personal meaning.
3. TACHS-R2-003 - Answer A: Careful attention can preserve objects and the memories they carry.
Maya's patient repair restores both the bell's function and its connection to Grandfather's past.
4. TACHS-R2-004 - Answer D: traveled gently.
The clear note moves softly through the room, so "floated" describes gentle travel.
5. TACHS-R2-005 - Answer A: Steady changes in air pressure caused by repeating swirls
Repeating air swirls produce regular pressure changes that people hear as a tone.
6. TACHS-R2-006 - Answer D: To explain an unfamiliar process through a familiar example
The bottle comparison helps readers picture how moving air across an opening can create sound.
7. TACHS-R2-007 - Answer B: It explains a cause, describes its effects, and presents responses.
The passage moves from how humming begins to why it matters and how engineers reduce it.
8. TACHS-R2-008 - Answer C: continuing.
A persistent tone is one that continues long enough to disturb residents or affect parts.
9. TACHS-R2-009 - Answer D: How empty areas contribute to observation and composition
The passage explains how negative space guides viewers, shapes mood, and helps artists observe proportions.
10. TACHS-R2-010 - Answer A: It can organize information and guide the viewer's eye.
Empty areas separate information and lead the eye through a poster's elements.
11. TACHS-R2-011 - Answer C: It provides an indirect way to judge the chair's proportions.
Correctly shaped gaps help the surrounding chair parts take on more accurate proportions.
12. TACHS-R2-012 - Answer B: explanatory and encouraging.
The author clearly explains the technique while emphasizing its usefulness to developing artists.
13. TACHS-R2-013 - Answer A: The relationship between elements can matter as much as the elements themselves.
The final paragraph stresses that objects and surrounding spaces work together to create a composition.
14. TACHS-R2-014 - Answer A: She tested the gravel path herself.
Nia rides the path to confirm that it is firm, wide, and safe before selecting it.
15. TACHS-R2-015 - Answer B: She feels responsible to the riders who prepared.
The passage immediately notes that several younger riders had practiced for weeks.
16. TACHS-R2-016 - Answer C: An unexpected obstacle can lead to a worthwhile discovery.
The blocked road leads the group to discover a garden that may become part of a future route.
17. TACHS-R2-017 - Answer D: solid enough to travel over.
"Firm" describes the gravel path as stable enough for the group to cross safely.
18. TACHS-R2-018 - Answer D: It slows the seed's downward motion.
Autorotation produces an upward force that reduces the seed's falling speed.
19. TACHS-R2-019 - Answer B: To explain how researchers observe and compare rapid motion
High-speed recordings let researchers examine how changed seed features affect rotation and descent.
20. TACHS-R2-020 - Answer A: It explains a natural process, describes how it is studied, and identifies a possible application.
The passage moves from seed flight to experiments and then to spinning sensor designs.
21. TACHS-R2-021 - Answer C: an example that can guide a design.
Engineers use the maple seed's simple motion-controlling shape as inspiration for devices.
22. TACHS-R2-022 - Answer C: Postal workers, firefighters, and delivery drivers
The second paragraph names these three groups as needing dependable property locations.

23. TACHS-R2-023 - Answer A: To explain why standardized addresses became necessary and how they improved navigation
The author traces the limitations of informal directions and the benefits of shared address rules.
24. TACHS-R2-024 - Answer B: Address systems replaced increasingly unreliable local directions with predictable, shared location information.
This choice captures both the problem caused by growth and the central function of standardized addresses.
25. TACHS-R2-025 - Answer D: A successful address system depends on consistent records as well as visible numbers.
Officials had to maintain records and resolve duplicate names in addition to placing signs and plaques.
26. TACHS-R2-026 - Answer A: She measured the space and checked the outlets.
Lena measures the hallway and counts working outlets before requesting permission.
27. TACHS-R2-027 - Answer B: They enjoy identifying details from their shared surroundings.
The neighbors slow down as they recognize familiar building details in the painted lanterns.
28. TACHS-R2-028 - Answer C: Creativity can preserve a tradition by giving it a new form.
Lena adapts the canceled courtyard walk to a basement display that still celebrates the community.
29. TACHS-R2-029 - Answer D: undecorated.
The hallway is described as visually simple before the neighbors add lanterns.
30. TACHS-R2-030 - Answer A: The seal around a screw failed.
A water trail reveals that the faulty screw seal allowed rain to escape.
31. TACHS-R2-031 - Answer B: It helps demonstrate the problem and the investigation that followed.
The leaking gauge provides visible evidence for discussing failure, repair, and scientific investigation.
32. TACHS-R2-032 - Answer C: approving of his honesty and willingness to learn.
The passage presents Owen's decision to disclose and study the failure as valuable scientific behavior.
33. TACHS-R2-033 - Answer D: information that helps explain what occurred.
The empty container provides information about the leak and therefore serves as evidence.
34. TACHS-R2-034 - Answer A: It can withstand the rain and dirt brought inside.
The passage directly states that entrance flooring may need durability because of rain and dirt.
35. TACHS-R2-035 - Answer B: How materials and room arrangement help manage sound
The passage explains both sound-absorbing materials and spatial planning used to control library noise.
36. TACHS-R2-036 - Answer C: It shifts from sound-absorbing materials to the placement of spaces and activities.
After discussing materials, the third paragraph explains how room arrangement separates noisy and quiet activities.
37. TACHS-R2-037 - Answer D: To emphasize that useful communication and safety sounds must remain audible
The author notes that communication, alarms, and a welcoming atmosphere require some sound.
38. TACHS-R2-038 - Answer A: Effective sound control depends on matching design choices to how each area is used.
The passage consistently connects materials and room placement with the particular needs of library spaces.
39. TACHS-R2-039 - Answer A: A note beside his name on the sign-in sheet
The note indicates that Mr. Ito is bringing supplies for the Tuesday group.
40. TACHS-R2-040 - Answer B: She wants evidence before deciding who owns it.
Elena follows clues but checks the record instead of treating familiarity as proof.
41. TACHS-R2-041 - Answer C: Patient investigation can reveal the meaning connected to an object.
Elena's careful search returns the apron and uncovers its connection to Mr. Ito's aunt.
42. TACHS-R2-042 - Answer D: determining the identity of.
Elena carefully determines who owns the apron before returning it.
43. TACHS-R2-043 - Answer A: The fraction of incoming sunlight a surface sends away
Paragraph 2 directly defines solar reflectance as the portion of incoming sunlight reflected.
44. TACHS-R2-044 - Answer B: To show that a roof's benefit varies with local conditions
Shade reduces direct sunlight, illustrating why the same roof change does not help every building equally.
45. TACHS-R2-045 - Answer C: It explains a technology, describes how it is measured, and discusses when it is useful.
The paragraphs move from the roof's function to product measurements, conditions, and combined planning.
46. TACHS-R2-046 - Answer D: a portion of a whole.
Solar reflectance is the portion of all incoming sunlight that the surface sends away.
47. TACHS-R2-047 - Answer B: Common street names and notes about access
The passage states that Bell's separate guide recorded local names and access information.

48. TACHS-R2-048 - Answer C: They benefited from information not shown by measurements alone.

Bell's access notes helped crews reach the fire by explaining how routes actually functioned.

49. TACHS-R2-049 - Answer D: To demonstrate how local knowledge can strengthen technical accuracy

Bell's interviews add practical information that makes her measured map more useful.

50. TACHS-R2-050 - Answer A: Bell produced a more useful map by combining measurements with knowledge of how people used the town.

This choice captures Bell's method and the passage's concluding principle.

Written Expression

51. TACHS-WE2-001 - Answer A: Sentence 3

The opening date is an isolated fact that interrupts the account of how students created the display.

52. TACHS-WE2-002 - Answer C: estimate when each photograph was taken

This revision states precisely what students inferred from the visual evidence.

53. TACHS-WE2-003 - Answer B: Finally,

Sentence 6 describes the final stage after research and photograph selection.

54. TACHS-WE2-004 - Answer D: After deciding to create a hallway display about the school building's history, our class searched old yearbooks and interviewed two former students.

The revision clearly subordinates the decision and identifies the class as the actor in both actions.

55. TACHS-WE2-005 - Answer B: Clear labels matter because they help gardeners choose seeds suited to their space.

"Because" correctly joins the claim to its reason without creating a comma splice.

56. TACHS-WE2-006 - Answer A: After sentence 6

The example of mold directly supports sentence 6's recommendation about proper seed storage.

57. TACHS-WE2-007 - Answer D: A volunteer could inspect the cabinet weekly and remove damaged packets.

This version preserves the meaning while removing unnecessary and repetitive wording.

58. TACHS-WE2-008 - Answer C: Requiring only modest supplies and regular care, the project could connect experienced gardeners with curious beginners.

"Modest supplies" and "regular care" form parallel noun phrases, and the modifier clearly describes the project.

59. TACHS-WE2-009 - Answer B: Neither the coach nor the players were expecting the field to be closed.

With "neither...nor," the verb agrees with the nearer plural subject "players."

60. TACHS-WE2-010 - Answer C: more confidently

The comparative adverb "more confidently" correctly modifies "spoke" when two finalists are compared.

61. TACHS-WE2-011 - Answer A: After Jordan placed the glass on the tray, he carried the tray to the table.

The sentence names each object and presents the actions in a clear sequence.

62. TACHS-WE2-012 - Answer D: Line 4

The preposition "between" requires the objective pronoun "me," not the subject pronoun "I."

63. TACHS-WE2-013 - Answer B: Line 4

The singular subject "Each" requires the singular verb "was placed."

64. TACHS-WE2-014 - Answer C: Sentence 3

The wall color is unrelated to the investigation of cafeteria-line delays and proposed improvements.

65. TACHS-WE2-015 - Answer A: Council members timed how long students spent in line during four lunch periods and recorded the locations of repeated delays.

This revision states exactly what was measured, when it was measured, and what was recorded.

66. TACHS-WE2-016 - Answer D: The manager tested both changes for one week, and the average wait became nearly two minutes shorter.

A comma and coordinating conjunction correctly join the two independent clauses.

67. TACHS-WE2-017 - Answer B: After sentence 4

"This pattern" refers directly to students waiting until the counter to read the menu.

68. TACHS-WE2-018 - Answer C: The packed sand made the bucket stable.

The revision removes redundancy and explains the sand's useful effect precisely.

69. TACHS-WE2-019 - Answer D: Although clouds sometimes made the shadow faint, each mark recorded the time and weather conditions whenever the shadow was visible.

The subordinating structure clearly connects the weather limitation with the recording process.

70. TACHS-WE2-020 - Answer A: On a fully cloudy morning, we could not make a reliable time mark.

The detail directly demonstrates the limitation of a timekeeping method that depends on sunlight.

71. TACHS-WE2-021 - Answer B: We began at nine in the morning, marking the tip of the stick's shadow every hour.

"Began" correctly maintains past tense, and "marking" properly describes the continuing action.

72. TACHS-WE2-022 - Answer C: The committee has submitted its recommendation.

The singular collective noun "committee" agrees with "has" and the singular pronoun "its."

73. TACHS-WE2-023 - Answer D: After the final bell rang, we packed our materials and left.
An introductory dependent clause requires a comma, while the compound predicate needs no internal comma.
74. TACHS-WE2-024 - Answer C: Line 3
The comparison requires “than,” not the time-related word “then.”
75. TACHS-WE2-025 - Answer A: Line 1
The adverb “quietly,” rather than the adjective “quiet,” must modify “slept.”
76. TACHS-WE2-026 - Answer B: Line 2
The surrounding past-tense narrative requires “checked,” not “checks.”
77. TACHS-WE2-027 - Answer A: Another team recorded the afternoon: buses stopped, shop doors opened, and basketballs struck the playground pavement.
The colon correctly introduces the complete list of afternoon sounds after an independent clause.
78. TACHS-WE2-028 - Answer B: We also removed clips overwhelmed by wind or unrelated conversation.
This detail logically extends the explanation of how clips were prepared before arrangement.
79. TACHS-WE2-029 - Answer C: In just eight minutes, the finished recording traces an entire day through its changing sounds.
The revision connects the recording’s length directly to its purpose and content.
80. TACHS-WE2-030 - Answer D: Because we labeled every clip by location and time, we could arrange the sounds from early morning through evening.
“Because” clearly shows that careful labeling enabled the chronological arrangement.
81. TACHS-WE2-031 - Answer A: On Saturday, our group painted the sign bright blue.
Active voice makes the sentence direct and consistent with the group’s actions elsewhere in the passage.
82. TACHS-WE2-032 - Answer B: Some letters and one arrow on the sign had faded completely.
Missing wording and direction explain why the older map was needed as a reference.
83. TACHS-WE2-033 - Answer C: After the paint dried, we applied a clear coating to shield the sign from moisture and sunlight.
This revision precisely identifies the environmental damage the coating is meant to prevent.
84. TACHS-WE2-034 - Answer D: The final decision belongs to Carla and him.
The object pronoun “him” correctly follows the preposition “to.”
85. TACHS-WE2-035 - Answer A: By the time the guests arrived, we had already arranged the chairs.
Past perfect correctly marks the arrangement as completed before the guests arrived.
86. TACHS-WE2-036 - Answer B: Mia photographed the leaves covered with frost beside the path.
“Covered with frost” is placed directly after “leaves,” the words it modifies.
87. TACHS-WE2-037 - Answer C: Line 3
The plural subject “members” requires “request,” not the singular verb “requests.”
88. TACHS-WE2-038 - Answer D: Line 4
The past-tense context requires “recycled,” not the present-tense form “recycle.”
89. TACHS-WE2-039 - Answer A: Sentence 3
The backpack detail does not contribute to the development or testing of the portable display.
90. TACHS-WE2-040 - Answer B: After sketching several designs, we built a small cardboard model of the strongest one, but it tipped when books were placed on one side.
This revision connects the actions clearly and avoids a fragment, comma splice, or misplaced modifier.
91. TACHS-WE2-041 - Answer C: In the second test, the display stayed balanced as two students carried it across the room.
The revision states exactly how the display was tested and what “steady” means.
92. TACHS-WE2-042 - Answer D: By testing and revising our design, we created a useful display that meets the club’s original needs.
The sentence concludes by connecting the finished product to the opening requirements.
93. TACHS-WE2-043 - Answer A: After reorganizing the questions into related groups, I added short follow-up questions to each group.
The introductory phrase clearly modifies “I” and establishes the sequence of actions.
94. TACHS-WE2-044 - Answer B: I found her explanation fascinating.
This revision removes redundant intensifiers while preserving the original meaning.
95. TACHS-WE2-045 - Answer C: For the school podcast, I interviewed Ms. Alvarez, who begins baking before sunrise.
The introductory phrase and nonrestrictive clause are both correctly set off with commas.

96. TACHS-WE2-046 - Answer A: One of the musicians practices in the auditorium each morning.
The singular subject "One" correctly takes the singular verb "practices."
97. TACHS-WE2-047 - Answer B: The science teacher spoke with Lena and me after class.
The object pronoun "me" correctly follows the preposition "with."
98. TACHS-WE2-048 - Answer C: Line 3
The surrounding past-tense sequence requires "thanked," not "thanks."
99. TACHS-WE2-049 - Answer D: Line 4
A comparison between two models requires the comparative form "easier," not "easiest."
100. TACHS-WE2-050 - Answer D: Line 4
The possessive determiner "their," not the location word "there," must modify "instruments."

Mathematics

101. TACHS-M2-001 - Answer B: 120 liters

The added 18 liters equals $\frac{3}{4} - \frac{3}{5} = \frac{3}{20}$ of capacity, so capacity is $18 \times 20 \div 3 = 120$.

102. TACHS-M2-002 - Answer D: 40

Multiplying both numbers by 100 gives $240 \div 6 = 40$.

103. TACHS-M2-003 - Answer A: \$72

The reduced price is $\$75 \times 0.80 = \60 , and increasing \$60 by 20% gives \$72.

104. TACHS-M2-004 - Answer C: 65

Each term is twice the preceding term minus 1, so $33 \times 2 - 1 = 65$.

105. TACHS-M2-005 - Answer B: $x \leq 7$

Add 7 to obtain $4x \leq 28$, then divide by 4 to get $x \leq 7$.

106. TACHS-M2-006 - Answer A: $3m + 5$

Six miles cost \$18, leaving a \$5 fixed fee; therefore the cost is $3m + 5$.

107. TACHS-M2-007 - Answer D: 86

Five scores total $5 \times 84 = 420$; the four known scores total 334, leaving 86.

108. TACHS-M2-008 - Answer C: $\frac{1}{36}$

Without replacement, the probability is $\frac{2}{9} \times \frac{1}{8} = \frac{2}{72} = \frac{1}{36}$.

109. TACHS-M2-009 - Answer A: 113°

Each obtuse angle is supplementary to a 67° acute angle, so its measure is $180^\circ - 67^\circ = 113^\circ$.

110. TACHS-M2-010 - Answer B: 54 square units

A right triangle's area is $\frac{1}{2} \times 9 \times 12 = 54$ square units.

111. TACHS-M2-011 - Answer C: 32 meters

Actual dimensions are 9 meters and 7 meters, so the perimeter is $2(9 + 7) = 32$ meters.

112. TACHS-M2-012 - Answer D: 2 hours 7 minutes

The full trip lasts 2 hours 29 minutes; subtracting the 22-minute stop gives 2 hours 7 minutes.

113. TACHS-M2-013 - Answer D: $1 \frac{11}{12}$ pounds

Convert to twelfths: $2 \frac{8}{12} - \frac{9}{12} = 1 \frac{11}{12}$ pounds.

114. TACHS-M2-014 - Answer A: 180

The least number containing the required prime factors 2^2 , 3^2 , and 5 is 180.

115. TACHS-M2-015 - Answer C: 35%

$\frac{7}{20} = \frac{35}{100}$, so 35% of the visitors bought a program.

116. TACHS-M2-016 - Answer B: 29

Substitute -4 : $y = 2(-4)^2 - 3 = 2(16) - 3 = 29$.

117. TACHS-M2-017 - Answer D: 25

The integers are 23, 24, and 25 because their sum is 72.

118. TACHS-M2-018 - Answer C: 21

In order, the values are 15, 18, 21, 24, 27, making 21 the middle value.

119. TACHS-M2-019 - Answer A: $\frac{9}{100}$

Replacement keeps both probabilities at $\frac{3}{10}$, so $\frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$.

120. TACHS-M2-020 - Answer D: 21

Five meetings must total 105; the first four total 84, so the fifth needs 21.

121. TACHS-M2-021 - Answer B: 8

The exterior angle is $180^\circ - 135^\circ = 45^\circ$, and $360^\circ \div 45^\circ = 8$ sides.

122. TACHS-M2-022 - Answer C: 100 cm^2

For a square, diagonal = $\text{side}\sqrt{2}$, so the side is 10 centimeters and the area is 100 cm^2 .

123. TACHS-M2-023 - Answer D: 158 cm^2

Surface area is $2(8 \times 5 + 8 \times 3 + 5 \times 3) = 2(79) = 158 \text{ cm}^2$.

124. TACHS-M2-024 - Answer A: 14

Convert 3.5 liters to 3,500 milliliters, then calculate $3,500 \div 250 = 14$.

125. TACHS-M2-025 - Answer B: 15
Convert 1 hour 48 minutes to 1.8 hours, then divide $27 \div 1.8 = 15$ miles per hour.
126. TACHS-M2-026 - Answer A: $\frac{2}{3}$
Dividing by $\frac{9}{10}$ means multiplying by $\frac{10}{9}$, so $\frac{3}{5} \times \frac{10}{9} = \frac{2}{3}$.
127. TACHS-M2-027 - Answer B: 25%
The decrease is \$16, and $16 \div 64 = 0.25 = 25\%$.
128. TACHS-M2-028 - Answer C: 21
Twenty-one divides 84 evenly and equals 7×3 .
129. TACHS-M2-029 - Answer D: Multiply by 2, then add 1.
Each term doubles and increases by 1: 5 becomes 11, then 23, then 47.
130. TACHS-M2-030 - Answer A: \$17.50
Substitute $m = 7$: $F = 3.50 + 2(7) = 17.50$.
131. TACHS-M2-031 - Answer B: 25
The consecutive integers are 23, 24, and 25, whose sum is 72.
132. TACHS-M2-032 - Answer C: 30
Six students need 150 total pages; the known five total 120, leaving 30.
133. TACHS-M2-033 - Answer D: $\frac{7}{8}$
Every whole number from 2 through 8 is prime or composite; only 1 is neither, giving $\frac{7}{8}$.
134. TACHS-M2-034 - Answer A: 71°
The base angles share $180^\circ - 38^\circ = 142^\circ$, so each measures 71° .
135. TACHS-M2-035 - Answer B: 44 cm
Circumference equals $\pi d = \frac{22}{7} \times 14 = 44$ centimeters.
136. TACHS-M2-036 - Answer C: 18 m
Multiply 7.2 centimeters by 2.5 meters per centimeter to obtain 18 meters.
137. TACHS-M2-037 - Answer D: 24
Convert 3.6 liters to 3,600 milliliters, then calculate $3,600 \div 150 = 24$.
138. TACHS-M2-038 - Answer A: 6 cups
Each batch uses $2 \frac{1}{4} \div 3 = \frac{3}{4}$ cup, and $8 \times \frac{3}{4} = 6$ cups.
139. TACHS-M2-039 - Answer C: 9
Evaluate parentheses first: $18 - 3 \times 6 \div 2 = 18 - 9 = 9$.
140. TACHS-M2-040 - Answer B: 80
If 115% equals 92, the original number is $92 \div 1.15 = 80$.
141. TACHS-M2-041 - Answer D: 9 cm
With width w and length $w + 5$, $2w + 2(w + 5) = 46$, so $w = 9$.
142. TACHS-M2-042 - Answer C: 7
Solve $4x - 7 = 21$: add 7 and divide by 4 to obtain $x = 7$.
143. TACHS-M2-043 - Answer A: 17
The range is the greatest value minus the least value: $35 - 18 = 17$.
144. TACHS-M2-044 - Answer D: $\frac{1}{2}$
Four of the eight sections are green or yellow, so the probability is $\frac{4}{8} = \frac{1}{2}$.
145. TACHS-M2-045 - Answer B: 34
Five days must total 150 passengers; the first four total 116, leaving 34.
146. TACHS-M2-046 - Answer C: 20
The scale factor is $15/6 = 2.5$, so the corresponding side is $8 \times 2.5 = 20$.
147. TACHS-M2-047 - Answer A: 5 cm
Since $31.4 = 2 \times 3.14 \times r$, the radius is 5 centimeters.
148. TACHS-M2-048 - Answer D: 4 inches
Height equals $180 \div (9 \times 5) = 4$ inches.
149. TACHS-M2-049 - Answer B: 8
Convert 14 feet to 168 inches and 1 foot 9 inches to 21 inches; $168 \div 21 = 8$.

150. TACHS-M2-050 - Answer C: 34 meters

Each centimeter represents 4 meters, so $8.5 \times 4 = 34$ meters.

Ability

151. TACHS-A2-001 - Answer C: ● ●

The bottom row must add one filled circle, matching the horizontal count rule and vertical fill change.

152. TACHS-A2-002 - Answer A: ←

Rotating the bottom-left arrow 90° clockwise gives left, while rotating the top-right arrow 180° also gives left.

153. TACHS-A2-003 - Answer D: Left ■ | Right ●

Filling the top-right cell gives ■ then ●, and swapping the bottom-left shapes produces the same arrangement.

154. TACHS-A2-004 - Answer B: ● ■

The third cell combines the first and second cells in their original left-to-right order.

155. TACHS-A2-005 - Answer C: △

Each row uses one shape in the pattern open, filled, open.

156. TACHS-A2-006 - Answer A: Near both upper corners

Unfolding across the vertical centerline reflects the punched hole to the opposite upper corner.

157. TACHS-A2-007 - Answer B: 4

Two perpendicular folds create four layers, so one central punch produces four reflected holes.

158. TACHS-A2-008 - Answer D: Two holes reflected across the diagonal, near the upper-right and lower-left corners

The diagonal fold reflects the off-fold punch to the corresponding position across that diagonal.

159. TACHS-A2-009 - Answer A: One complete circular hole centered on the horizontal centerline

The semicircular cut reflects across the folded edge to form one complete circle.

160. TACHS-A2-010 - Answer C: ◇ ◆ ◇

Each group member has matching open outer shapes and the filled version of that shape at the center.

161. TACHS-A2-011 - Answer D: ← ↑ →

In every group member, each successive arrow rotates 90° clockwise.

162. TACHS-A2-012 - Answer B: Outer ◇ | Center ★ | Outer ◇

Each arrangement has identical open outer shapes surrounding one different filled center shape.

163. TACHS-A2-013 - Answer B: ◇

Removing ★ and + from the first cell leaves only ◇ in its original position.

164. TACHS-A2-014 - Answer D: Arrow ↑ inside a filled circle ●

Either stated transformation produces an upward arrow inside a filled circle.

165. TACHS-A2-015 - Answer A: □

A square completes the one-of-each-shape pattern in both the last row and last column.

166. TACHS-A2-016 - Answer C: Left ◆ | Right ▲

Reflection places ◇ left of △, and filling them changes the pair to ◆ left of ▲.

167. TACHS-A2-017 - Answer A: Four holes, one near each corner

The two perpendicular center folds reflect the punch across both centerlines, producing four corner positions.

168. TACHS-A2-018 - Answer C: Two matching triangular notches on the left edge, mirrored across the horizontal centerline

Unfolding reflects the outer-edge cut vertically while keeping both notches on the left boundary.

169. TACHS-A2-019 - Answer D: 4

Two folds create four layers at the punch, so reversing both folds produces four symmetric holes.

170. TACHS-A2-020 - Answer B: Two matching semicircular notches along the top edge

The vertical fold reflects the top-edge cut to the corresponding position on the other half.

171. TACHS-A2-021 - Answer C: ◇ above ◆

Each valid pair places an open shape above its filled counterpart.

172. TACHS-A2-022 - Answer B: ← ◇ →

In each valid arrangement, the outer arrows point in opposite directions with an open shape between them.

173. TACHS-A2-023 - Answer A: Open triangle △ containing +

Each valid figure is an open boundary shape containing a plus sign.

174. TACHS-A2-024 - Answer D: □ △ □

Each valid arrangement has identical open outer shapes surrounding a different open center shape.

175. TACHS-A2-025 - Answer B: $\triangle \blacktriangle$
Each third cell joins the open shape from the first cell to the filled shape from the second.
176. TACHS-A2-026 - Answer A: $\uparrow$
Rotating $\downarrow$ clockwise gives $\leftarrow$, and rotating $\leftarrow$ clockwise gives $\uparrow$.
177. TACHS-A2-027 - Answer C: $\star$
The star is the only shape shared by the first two cells in the third row.
178. TACHS-A2-028 - Answer D: Bottom-left $\bullet$; other corners $\circ$
The filled circle moves clockwise from top-left to top-right to bottom-right and then bottom-left.
179. TACHS-A2-029 - Answer B: $\blacksquare \star$
Changing $\square$ from open to filled gives $\blacksquare$, while changing $\star$ from filled to open gives $\star$.
180. TACHS-A2-030 - Answer A: 4
Two independent folds create four layers, so one punch away from folds produces four reflected holes.
181. TACHS-A2-031 - Answer C: Four holes forming two left-right mirror pairs
Each of the two punches reflects once across the vertical fold, producing two distinct mirrored pairs.
182. TACHS-A2-032 - Answer D: One diamond-shaped hole centered on the diagonal
A triangular cut on the folded diagonal reflects across that fold to form one diamond-shaped opening.
183. TACHS-A2-033 - Answer B: One cut at the center of the original sheet
The described packet corner lies at the intersection of both folds, so unfolding leaves one central cut.
184. TACHS-A2-034 - Answer A: $\circ$ inside $\triangle$
The inner circle has no straight sides, fewer than the outer triangle's three sides.
185. TACHS-A2-035 - Answer C: $\diamond \blacklozenge \diamond$
The two outside shapes are matching open diamonds, and the center is the filled form of the diamond.
186. TACHS-A2-036 - Answer D: Left: $\leftarrow$; Right: $\rightarrow$
Both arrows point away from the center along the horizontal line joining their positions.
187. TACHS-A2-037 - Answer B: $\triangle | \square | \circ \circ \circ$
The counts increase from one to three, and each position uses a different shape.
188. TACHS-A2-038 - Answer D: $\diamond \bullet$
Symbols appearing in both cells cancel, leaving the diamond from the first cell and circle from the second.
189. TACHS-A2-039 - Answer A: $\uparrow$ above $\circ$
Moving right rotates the arrow clockwise and switches the shape's fill in both rows.
190. TACHS-A2-040 - Answer B: bottom-left $\star$
The star moves clockwise through the corners while alternating open, filled, open, filled.
191. TACHS-A2-041 - Answer C: $\bullet \star$
Reversing $\star \circ$ gives $\circ \star$, and filling both shapes produces $\bullet \star$.
192. TACHS-A2-042 - Answer D: Two holes reflected across the diagonal, near the upper-right and lower-left corners
Unfolding reflects the off-fold punch across the named diagonal.
193. TACHS-A2-043 - Answer B: Four matching corner cuts, one at each corner
Two center folds bring all four original corners together, so the cut reflects to every corner.
194. TACHS-A2-044 - Answer A: One complete circular hole left of center on the horizontal centerline
The semicircular notch lies on the fold and reflects into its matching half when unfolded.
195. TACHS-A2-045 - Answer D: Four holes, forming upper and lower pairs reflected across the vertical centerline
Each of the two punches reflects once across the vertical fold, producing four holes.
196. TACHS-A2-046 - Answer B: $\diamond | \blacklozenge$
Each arrangement pairs one open shape with two filled versions of the same shape.
197. TACHS-A2-047 - Answer C: $\rightarrow \diamond \leftarrow$
In each valid arrangement, the two arrows point inward toward the central open shape.
198. TACHS-A2-048 - Answer D: Outer $\star$ containing inner $\diamond$
The inner shape must match the previous figure's outer diamond, while the new outer shape changes.
199. TACHS-A2-049 - Answer C: Top: $\diamond$ | Center: $|$ | Bottom: $\diamond$
Each valid arrangement has identical open shapes above and below a different line symbol.

200. TACHS-A2-050 - Answer A: ◇ ◆ ◇ ◆

The sequence alternates the open and filled versions of one shape, beginning with the open form.